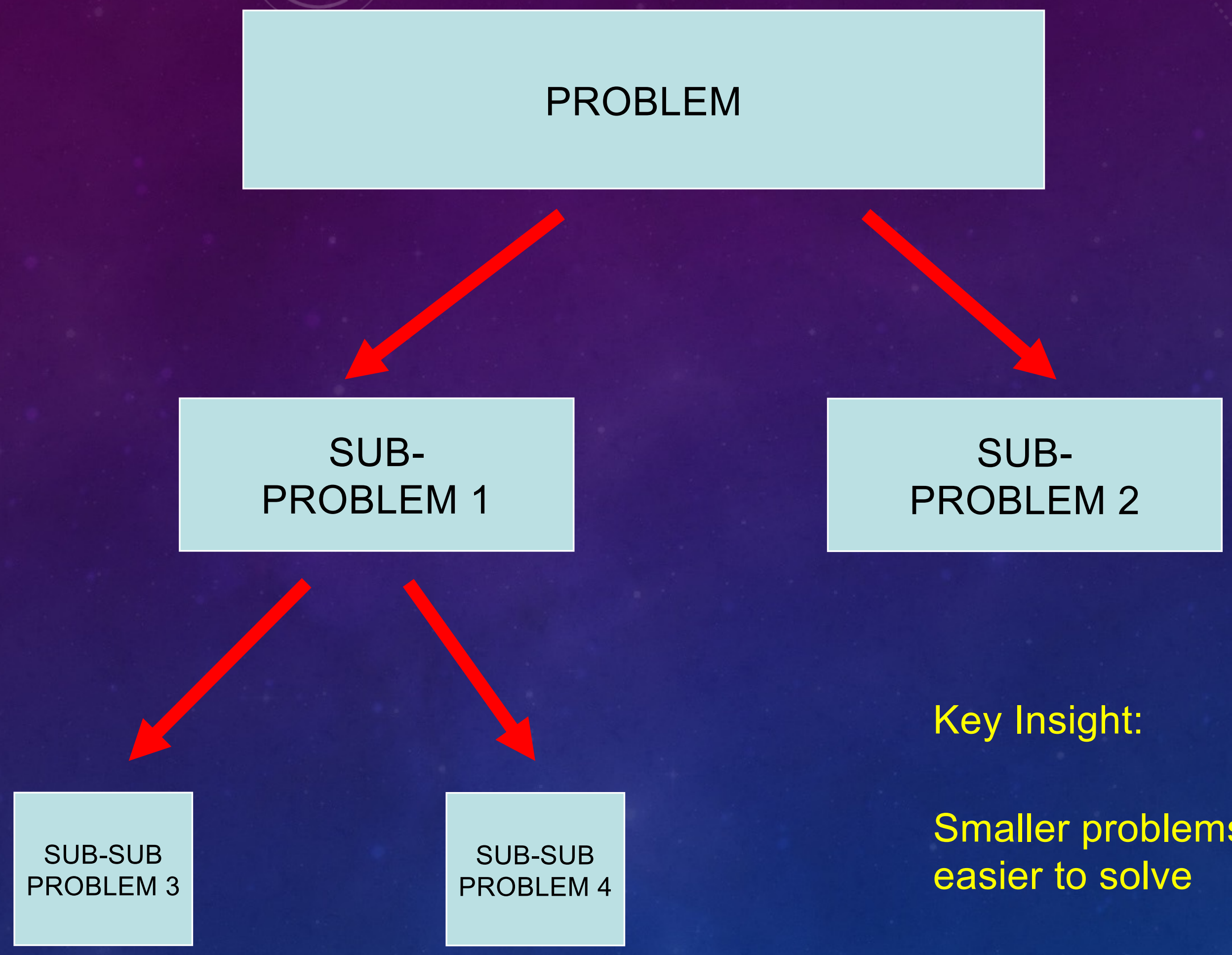
LECTURE 3B - GOOD ABSTRACTION

LT3B OBJECTIVES

- Know what abstraction means and understand its advantages
- Understand key approaches to problem solving

MANAGING COMPLEXITY

DIVIDE-AND-CONQUER



Key Insight:

Smaller problems are easier to solve

Abstractions

Problem

Primitives

Abstractions

Solution

Invented to make task easier

What makes a good abstraction?

Good Abstraction

1. Makes it more natural to think about tasks and subtasks
2. Makes programs easier to understand
3. Captures common patterns
4. Allows for the code to be reused
5. Hides the implementation details
6. Separates specification from implementation
7. Makes debugging (fixing errors) easier

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Example

House $\cancel{\longrightarrow}$ Bricks?

House

$\downarrow$

Rooms

$\downarrow$

Bricks

Divide and Conquer

Programming

Program $\cancel{\longrightarrow}$ Primitives

Program

$\downarrow$

Functions

$\downarrow$

Primitives

Divide and Conquer

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Compare:

```
def sum_of_squares(x, y):
    return square(x) + square(y)

def square(x):
    return x * x

def hypotenuse(a, b):
    return sqrt(sum_of_squares(a, b))
```

Versus:

```
def hypotenuse(a, b):
    return sqrt((a*a) + (b*b))
```

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Item	Amount ($)
Food	300
Entertainment	50
Saving	400
Others	250

```
my_finances =
    'food = $' + str(1000*0.3)\
    + '; entertainment = $' + str(1000*0.05)\
    + '; savings = $' + str(1000*0.4)\
    + '; others = $' + str(1000*0.25)
```

```
my_finances =
    'food = $' + str(1000*0.3)\
    + '; entertainment = $' + str(1000*0.05)\
    + '; savings = $' + str(1000*0.4)\
    + '; others = $' + str(1000*0.25)
```

Using Function to capture the common patterns

```
def my_finances(salary):
    return 'food = $' + str(salary*0.3)\
    + '; entertainment = $' + str(salary*0.05)\
    + '; savings = $' + str(salary*0.4)\
    + '; others = $' + str(salary*0.25))
```

```
>>> my_finances(1000)
```

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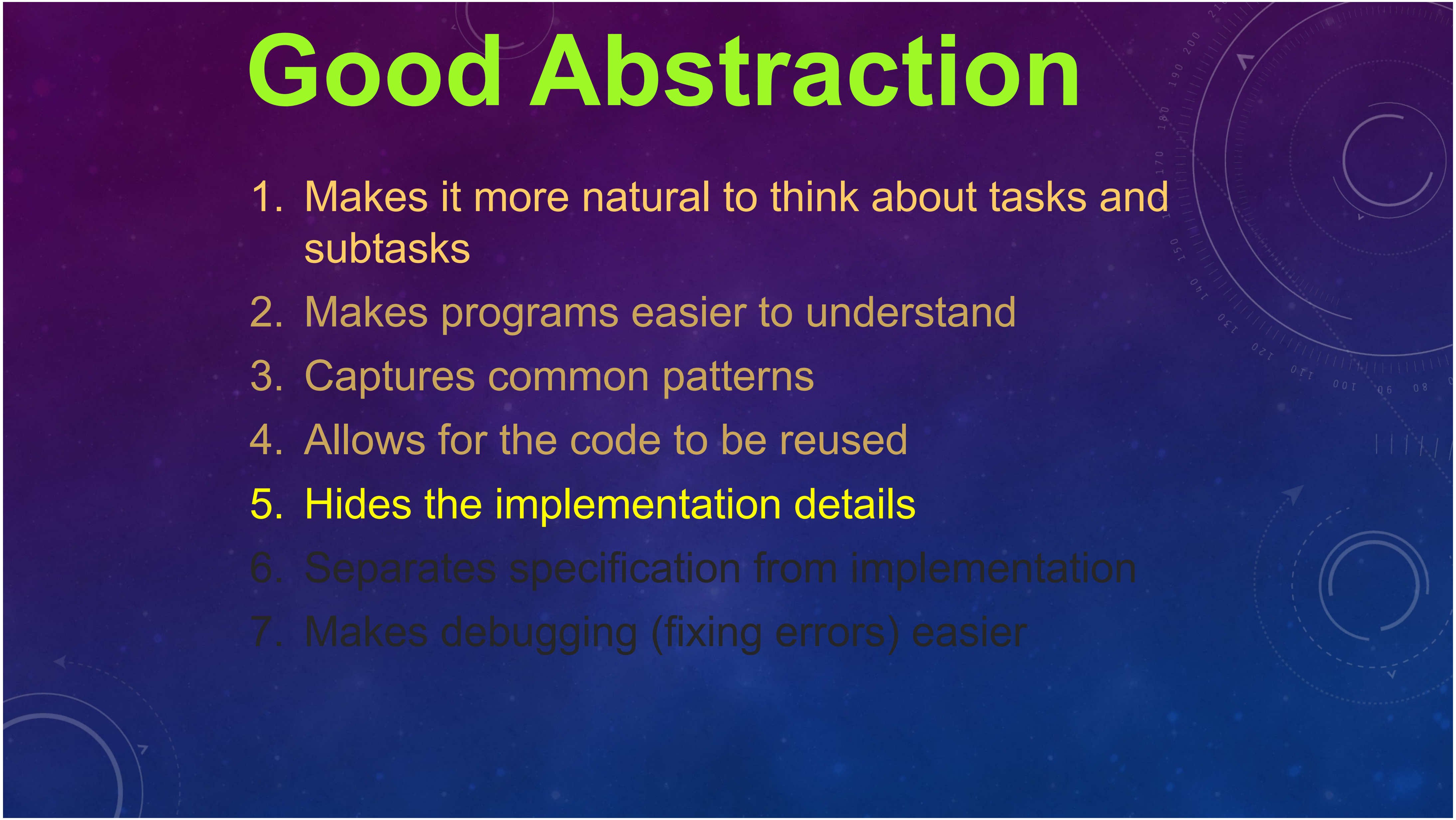
```
def square(x):
    return x * x
```

Function `square` used to find the hypothenuse can also be used to calculate the area of a circle.

Function to calculate area of circle, given its radius:

```
pi = 3.14159
def area_of_circle(r):
    return pi * square(r)
```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Input
(radius)

Area of Circle
Generator

Output
(Area of Circle)

provided
function

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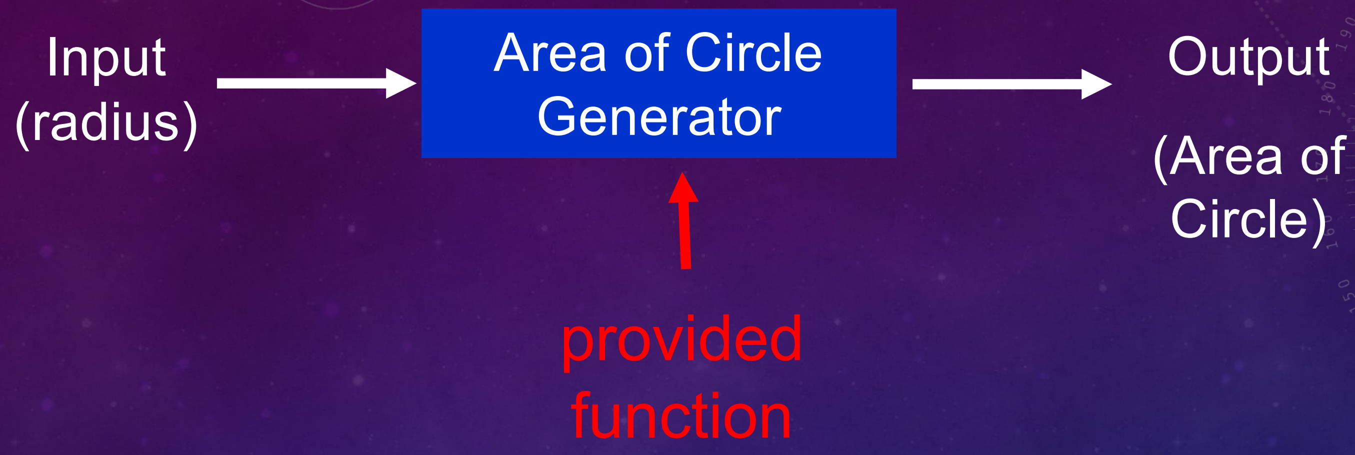
Recap:
Functional Abstraction
=
Black Box

No need to know how a car works to drive it!

Functional Abstraction: Separates specification from implementation

Specification: **WHAT** it should do

Implementation: **HOW** it does it



Specification: WHAT

(Calculate the area of a circle given its radius)

Implementation: HOW

(inside the box)

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Where is the bug?
Which is easier to check?

Code #1:

```
def hypotenuse(a, b):
    return sqrt((a + a) * (b + b))
```

Where is the bug?
Which is easier to check?

Code #2:

```
def square(x):
     return x + x

def sum_of_squares(x, y):
     return square(x) + square(y)

def hypotenuse(a,b):
     return sqrt(sum_of_squares(a,b))
```

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Wishful Thinking

Top-down approach:
Pretend you have whatever you need (!)

An Example

Singapore Taxi determines the taxi fare based on distance traveled as follows:

For the first kilometre or less: $3.00

Every 400 metres thereafter or less up to 10 km: $0.22

Every 400 metres thereafter or less after 10 km: $0.25

Problem: **Write a Python function that computes the taxi fare from the distance traveled.**

How do we start?

Formulate the problem

Function

Needs a name!

Pick an appropriate name

Formulate the problem

distance → `taxi_fare` → **fare**

- What data do you need? (be thorough)
- Where would you get it? (argument/ computed?)
- Results should be unambiguous

What other abstractions may be useful?

Ask the same questions for each abstraction.

How can the result be computed from data?

1. Try simple examples.
2. Strategize step by step.
3. Write it down and refine.

Solution

- **What to call the function?** `taxi_fare`
- **What data are required?** `distance`
- **How to get data?** `use an argument`
- **What is the result?** `fare`

HOW CAN THE RESULT BE COMPUTED FROM DATA?

- e.g.#1: distance = 800 m, fare = \$3.00

- e.g.#2: distance = 3300 m,

  fare = \$3.00 +roundup(2300/400) × \$0.22 = \$4.32

- e.g.#3: distance = 14500 m,

  fare = \$3.00 +roundup(9000/400) × \$0.22 + roundup(4500/400)× \$0.25 = \$11.06

Pseudocode

Case 1: distance <= 1000

fare = $3.00

Case 2: 1000 < distance <= 10,000

fare = $3.00 + $0.22 *

roundup((distance – 1000)/ 400)

Case 3: distance > 10,000 What's this? fare for 10 km

fare = $8.06 + $0.25 *

roundup((distance – 10,000)/ 400)

Note: the Python function `ceil` rounds up its argument. `math.ceil(1.5) = 2`

Solution

Remember to import `math` library to use `ceil`

```
from math import *
def taxi_fare(distance): # distance in metres
  if distance <= 1000:
    return 3.0
  elif distance <= 10000:
    return 3.0 + (0.22 *
                  ceil((distance - 1000) / 400))
  else:
    return 8.06 + (0.25 *
                  ceil((distance - 10000) / 400))

# check: taxi_fare(3300) = 4.32
```

Can we improve this solution?

Coping with Change

1. What if starting fare is increased to $3.20?

2. What if 400 m is decreased to 300 m?

Avoid Magic Numbers

- It is a **terrible idea** to hardcode constants (**magic numbers**):
  - –Hard to make changes in future.

Which are the magic numbers?

```
from math import *
def taxi_fare(distance): # distance in metres
  if distance <= 1000:
    return 3.0
  elif distance <= 10000:
    return 3.0 + (0.22 *
                    ceil((distance - 1000) / 400))
  else:
    return 8.06 + (0.25 *
                    ceil((distance - 10000) / 400))
```

How to replace the magic numbers?

```
from math import *
def taxi_fare(distance): # distance in metres
  start_fare = 3.0
  stage = 400

  if distance <= 1000:
    return start_fare
  elif distance <= 10000:
    return start_fare + (0.22 *
                ceil((distance - 1000) / stage))
  else:
    return 8.06 + (0.25 *
                ceil((distance - 10000) / stage))
```

How to remove this one???

```
taxi_fare(10000)
```

```
from math import *
def taxi_fare(distance): # distance in metres
    start_fare = 3.0
    block = 400

    if distance <= 1000:
        return start_fare
    elif distance <= 10000:
        return start_fare + (0.22 *
                       ceil((distance - 1000) / block))
    else:
        return taxi_fare(10000) + (0.25 *
                       ceil((distance - 10000) / block))
```

Easily cope with changes

1. Increase the starting fare to $3.20

2. Decrease the 400 m block to 300 m

Can easily update these magic numbers

```
from math import *
def taxi_fare(distance): # distance in metres
    start_fare = 3.2
    block = 300

    if distance <= 1000:
        return start_fare
    elif distance <= 10000:
        return start_fare + (0.22 *
                    ceil((distance - 1000) / block))
    else:
        return taxi_fare(10000) + (0.25 *
                    ceil((distance - 10000) / block))
```

Remove all the magic numbers !

```
from math import *
def taxi_fare(distance): # distance in metres
   stage1 = 1000
   stage2 = 10000
   start_fare = 3.0
   increment1 = 0.22
   increment2 = 0.25
   block = 400

   if distance <= stage1:
       return start_fare
   elif distance <= stage2:
       return start_fare + (increment1 *
            ceil((distance - stage1) / block))
   else:
       return taxi_fare(stage2) +
           (increment2 * ceil((distance - stage2) / block))
```